

Practice Problems: Elasticity Using Derivatives

GSE 510

Reminder. If $y = f(x)$, then the elasticity of y with respect to x is

$$\varepsilon_{yx} = \frac{dy}{dx} \cdot \frac{x}{y}.$$

This measures the approximate percentage change in y caused by a one percent change in x .

1. Linear Demand Elasticity

Suppose demand is given by

$$Q(p) = 100 - 2p.$$

(a) Find $\frac{dQ}{dp}$.

(b) Find the price elasticity of demand:

$$\varepsilon_{Qp} = \frac{dQ}{dp} \cdot \frac{p}{Q}.$$

(c) Evaluate the elasticity at $p = 20$.

(d) Interpret your answer.

Solution.

(a) Since

$$Q(p) = 100 - 2p,$$

we have

$$\frac{dQ}{dp} = -2.$$

(b) The elasticity formula is

$$\varepsilon_{Qp} = \frac{dQ}{dp} \cdot \frac{p}{Q}.$$

Substituting $\frac{dQ}{dp} = -2$, we get

$$\varepsilon_{Qp} = -2 \cdot \frac{p}{Q}.$$

Since $Q = 100 - 2p$,

$$\varepsilon_{Qp} = -2 \cdot \frac{p}{100 - 2p}.$$

(c) At $p = 20$,

$$Q(20) = 100 - 2(20) = 60.$$

Therefore,

$$\varepsilon_{Qp} = -2 \cdot \frac{20}{60} = -\frac{40}{60} = -\frac{2}{3}.$$

(d) At $p = 20$, the price elasticity of demand is

$$-\frac{2}{3}.$$

This means that a one percent increase in price is associated with approximately a $\frac{2}{3}$ percent decrease in quantity demanded. Since the absolute value is less than 1, demand is inelastic at this price.

2. Constant Elasticity Demand

Suppose demand is given by

$$Q(p) = 50p^{-2}.$$

- (a) Find $\frac{dQ}{dp}$.
- (b) Find the price elasticity of demand.
- (c) Does elasticity depend on p ?
- (d) Interpret the elasticity.

Solution.

- (a) Since

$$Q(p) = 50p^{-2},$$

we have

$$\frac{dQ}{dp} = 50(-2)p^{-3} = -100p^{-3}.$$

- (b) The elasticity is

$$\varepsilon_{Qp} = \frac{dQ}{dp} \cdot \frac{p}{Q}.$$

Substituting,

$$\varepsilon_{Qp} = (-100p^{-3}) \cdot \frac{p}{50p^{-2}}.$$

Simplify:

$$\varepsilon_{Qp} = \frac{-100p^{-2}}{50p^{-2}} = -2.$$

- (c) No. The elasticity does not depend on p . It is constant and equal to

$$\varepsilon_{Qp} = -2.$$

- (d) This means that a one percent increase in price leads to an approximate two percent decrease in quantity demanded. Since the absolute value is greater than 1, demand is elastic.

3. Income Elasticity of Demand

Suppose demand for a good depends on income m :

$$x(m) = 5 + 2m^{1/2}.$$

- (a) Find $\frac{dx}{dm}$.
- (b) Find the income elasticity of demand:

$$\varepsilon_{xm} = \frac{dx}{dm} \cdot \frac{m}{x}.$$

- (c) Evaluate the income elasticity at $m = 100$.
- (d) Is the good normal or inferior at $m = 100$?

Solution.

- (a) Since

$$x(m) = 5 + 2m^{1/2},$$

we have

$$\frac{dx}{dm} = 2 \cdot \frac{1}{2}m^{-1/2} = m^{-1/2}.$$

(b) The income elasticity is

$$\varepsilon_{xm} = \frac{dx}{dm} \cdot \frac{m}{x}.$$

Substituting,

$$\varepsilon_{xm} = m^{-1/2} \cdot \frac{m}{5 + 2m^{1/2}}.$$

Since

$$m^{-1/2}m = m^{1/2},$$

we get

$$\varepsilon_{xm} = \frac{m^{1/2}}{5 + 2m^{1/2}}.$$

(c) At $m = 100$,

$$x(100) = 5 + 2(10) = 25.$$

Also,

$$\left. \frac{dx}{dm} \right|_{m=100} = \frac{1}{10}.$$

Therefore,

$$\varepsilon_{xm} = \frac{1}{10} \cdot \frac{100}{25} = \frac{1}{10} \cdot 4 = 0.4.$$

(d) Since the income elasticity is positive, the good is normal at $m = 100$. A one percent increase in income increases demand by approximately 0.4 percent.